Introduction to Biophysics II: Quantum physics & biology

Lecture 17:
Quantum coherence in biology - 2

Reading:
Modern optical spectroscopy, chapter 10

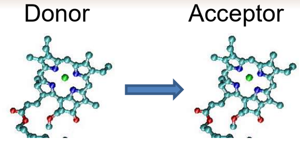
1

Coherence: oscillations

Last time…

Hamiltonian: $\tilde{\mathrm{H}} = \tilde{\mathrm{H}}_1 + \tilde{\mathrm{H}}_2 + \tilde{\mathrm{H}}_{21}$ *interaction*

Supermolecular state:



Excitation on donor
Excitation on acceptor

$$\Psi_B = C_1\psi_1 + C_2\psi_2 = C_1\phi_{1b}\chi_{1b}\phi_{2a}\chi_{2a} + C_2\phi_{1a}\chi_{1a}\phi_{2b}\chi_{2b}$$

Solve time-dependent SE: $\hat{H}\Psi = i\hbar\frac{\partial\Psi}{\partial t}$

Solution for $C_1(0)=1$, $C_2(0)=0$:

Population dynamics:

$$|C_2(t)|^2 = (1/2)(H_{21}/\Omega)^2[1 - \cos(2\Omega t/\hbar)] \qquad E_{21} = H_{22} - H_{11}$$

$$\Omega = (1/2)\left[(E_{21})^2 + 4(H_{21})^2\right]^{1/2}$$

- Population on acceptor oscillates with period $T = 2\pi\hbar/2\Omega = h/2\Omega$
- Mean probability is $(H_{21}/\Omega)^2/2$

Note: Ω is exactly the same as energy splitting in excitonic states (Lecture 15): $\pm\frac{1}{2}\sqrt{\delta^2 + 4(H_{12})^2}$

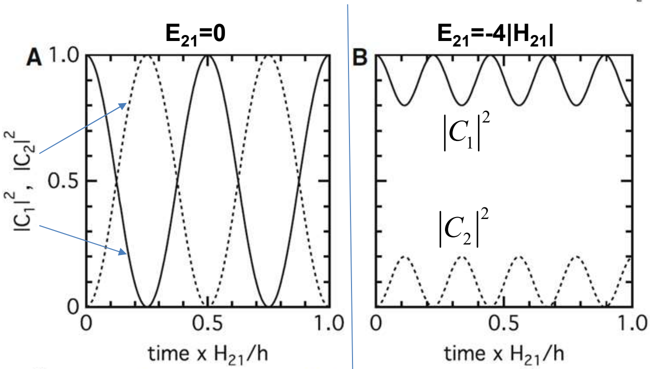
2

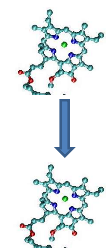
Population oscillations

Last time…

Special cases: ($C_1(0)=1$, $C_2(0)=0$)

$$\Omega = (1/2)\left[(E_{21})^2 + 4(H_{21})^2\right]^{1/2}$$





$$|C_2(t)|^2 = (1/2)[1 - \cos(2H_{21}t/\hbar)]$$

$T = \hbar/2\Omega = \hbar/(2H_{21})$

Even with very small H_{21} the system will be entirely in state 2 at some times!
Larger H_{21} increases frequency

$$|C_2(t)|^2 = 2(H_{21}/E_{21})^2[1 - \cos(E_{21}t/\hbar)]$$

$T = \hbar/2\Omega \approx \hbar/E_{21}$

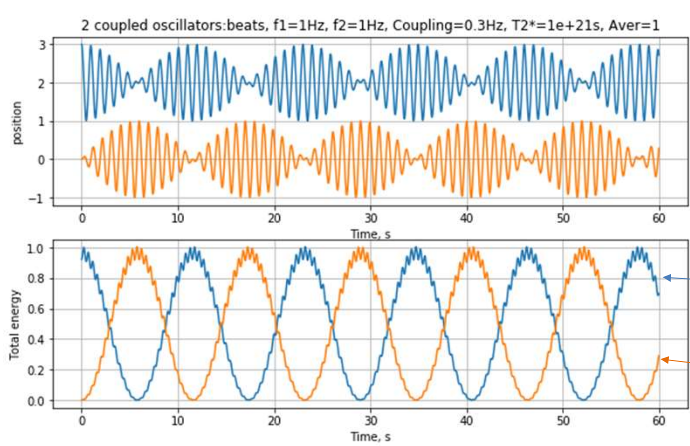
Larger E_{21} :
- Frequency increases
- Amplitude decreases
- The probability of transfer decreases

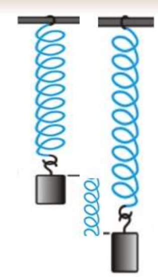
3

Classical analog: spring-mass oscillators

Last time…

Simulation using a simple Python script:
T=0: spring 1 (blue) is stretched, spring 2 (orange) is at equilibrium
Resonance frequencies: $\omega_1=\omega_2$





$$\ddot{x}_1 = -\omega_1^2 x_1 - \omega_{c1}^2 x_{12}$$
$$\ddot{x}_2 = -\omega_2^2 x_2 - \omega_{c2}^2 x_{21}$$

$|C_1|^2$
$$E_1 = \frac{m_1 v_1^2}{2} + \frac{k_1 x_1^2}{2}$$

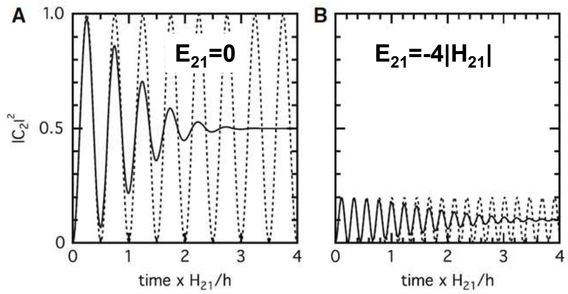
$|C_2|^2$
$$E_2 = \frac{m_2 v_2^2}{2} + \frac{k_2 x_2^2}{2}$$

Q: Why there are fast oscillations in both energies?

4

Last time…

Population oscillations: damping



1. **Dephasing due to ensemble inhomogeneity**
2. **Coupling with bath**
3. **Finite lifetimes of excited states**

5

The density matrix

Consider an isolated two-state system with the following wavefunction:

$$\Psi(t) = C_1(t)\boxed{\psi_1 exp[-i(H_{11}t/\hbar + \zeta_1)]} + C_2(t)\psi_2 exp[-i(H_{22}t/\hbar + \zeta_2)]$$

Split wavefunction into spatial and time-dependent parts

ψ_1, ψ_2 – spatial wavefunctions of two basis states with energies H_{11} and H_{22}
ζ_1, ζ_2 – phase shift, which depends on the time when state 1 or 2 is created

Let us redefine C coefficients: $c_n(t) = C_n(t)\exp[-i(H_{nn}t/\hbar + \zeta_n)]$

Then the wavefunction can be rewritten as: (combined C and time-dependent part of wavefunction)

$$\Psi(t) = c_1(t)\psi_1 + c_2(t)\psi_2$$

As usual, we can find the expectation value of any dynamic property *A*:

$$\begin{aligned}\langle A(t)\rangle &= \left\langle \Psi(t)\middle|\widetilde{\mathrm{A}}\middle|\Psi(t)\right\rangle = \left\langle c_1(t)\psi_1 + c_2(t)\psi_2\middle|\widetilde{\mathrm{A}}\middle|c_1(t)\psi_1 + c_2(t)\psi_2\right\rangle \\ &= c_1^*(t)\,c_1(t)\left\langle\psi_1\middle|\widetilde{\mathrm{A}}\middle|\psi_1\right\rangle + c_1^*(t)\,c_2(t)\left\langle\psi_1\middle|\widetilde{\mathrm{A}}\middle|\psi_2\right\rangle + c_2^*(t)c_1(t)\left\langle\psi_2\middle|\widetilde{\mathrm{A}}\middle|\psi_1\right\rangle \\ &+ c_2^*(t)\,c_2(t)\left\langle\psi_2\middle|\widetilde{\mathrm{A}}\middle|\psi_2\right\rangle \\ &= c_{\cancel{2}\,1}^*(t)c_1(t)A_{11} + c_1^*(t)c_2(t)A_{12} + c_2^*(t)c_1(t)A_{21} + c_2^*(t)c_2(t)A_{22},\end{aligned}$$

6

The density matrix

More generally, for any system that can be described by a linear combination of basis states, the expectation value *A*:

$$\Psi(t) = c_1(t)\psi_1 + c_2(t)\psi_2 + \ldots$$
$$c_n(t) = C_n(t)\exp[-i(H_{nn}t/\hbar + \zeta_n)]$$

$$\langle A(t)\rangle = \sum_m \sum_n c_m^*(t)c_n(t)\left\langle \psi_m \middle| \tilde{A} \middle| \psi_n \right\rangle = \sum_m \sum_n c_m^*(t)c_n(t)A_{mn}$$

Basis *spatial* wavefunctions (no time-dependence)

All the observable time dependence of the system thus resides in the array of products of the coefficients $c_m^* c_n$

Introduce **density matrix**:

$$\boxed{\rho_{nm}(t) = c_n(t)c_m^*(t)}$$

This is product of two matrices $= (\rho\mathbf{A})_{nn}$

Rewrite:

$$\langle A(t)\rangle = \sum_n \sum_m c_n(t)c_m^*(t)A_{mn} = \sum_n \sum_m \rho_{nm}(t)A_{mn}$$

$$\boxed{\langle A(t)\rangle = \sum_n (\rho A)_{nn} = Tr(\boldsymbol{\rho}\mathbf{A})}$$

Tr ≡ trace of a matrix, sum of all diagonal elements

7

The density matrix

The density matrix approach is very general:

If a wave function can be expressed as a sum of basis spatial wave functions

$$\Psi(\vec{r},t) = \sum_i c_i(t)\psi_i(\vec{r})$$

The expectation value of any physical quantity *A* can be found as

$$\langle A(t)\rangle = Tr(\boldsymbol{\rho}\mathbf{A}) \qquad \langle A(t)\rangle_{ensemble} = Tr\big(\boldsymbol{\rho}(t)\mathbf{A}\big)$$

Average over ensemble

With the density matrix..

$$\rho_{nm}(t) = c_n(t)c_m^*(t)$$

And the corresponding operator representing quantity *A* (in matrix form, same basis!):

$$A_{mn} = \left\langle \psi_m(\vec{r}) \,|\, \tilde{A} \,|\, \psi_n(\vec{r}) \right\rangle$$

NOTE:

*This is **Schrödinger representation** of density matrix, where* $c_n(t) = C_n(t)e^{-iH_{nn}t/\hbar}$

Interaction representation $\rho_{nm} = C_n(t)C_m^*(t)$, *introduce* $e^{-iH_{nn}t/\hbar}$ *separately*

8

The density matrix

$$\boxed{\rho_{nm}(t) = c_n(t)c_m^*(t)}$$

$$c_n(t) = C_n(t)e^{-iH_{nn}t/\hbar}$$

Notes:

- **Diagonal** terms of ρ loose exponentials (oscillations cancel):

$$\rho_{nn}(t) = \left[C_n^*(t)e^{iH_{nn}t/\hbar}\right]\left[C_n(t)e^{-iH_{nn}t/\hbar}\right] = C_n^*(t)C_n(t) = |C_n(t)|^2$$

- When ρ is averaged over the ensemble, diagonal elements (ρ_{nn}) can be interpreted as populations of corresponding basis states

- **Off-diagonal** elements can be complex and generally oscillate at the frequency defined by the energy difference of basis states

$$\rho_{nm} = C_n(t)C_m^*(t)\exp(-iE_{nm}t/\hbar)\exp(-i\zeta_{nm})$$

$$\text{with } E_{nm} = E_n - E_m,\ \zeta_{nm} = \zeta_n - \zeta_m,\ \text{and } n \neq m.$$

- The magnitudes of the ensemble-averaged off-diagonal elements provide information on the coherence of the ensemble
- The average over time should be zero

9

The density matrix

Example:

$$\boxed{\rho_{nm}(t) = c_n(t)c_m^*(t)}$$

N molecules (systems), each has two states, 1 and 2; initially, all are in state 1:

$C_{1(k)} = 1$, $C_{2(k)} = 0$ for k =1,…,N

Subject the ensemble to a short light pulse, any given molecule k can be promoted to the excited state with a certain probability:

$C_{2(k)} \neq 0$

The probability of finding a molecule in the ensemble in state 2 is the diagonal element

$$\overline{\rho_{22}}(t) = N^{-1}\sum_{k=1}^{N} c_{2(k)}(t)c_{2(k)}{}^*(t) = |\overline{c_2}(t)|^2 = |\overline{C_2}(t)|^2$$

$$c_n(t) = C_n(t)e^{-iH_{nn}t/\hbar}$$

The off-diagonal element:

$$\rho_{12}(t) = N^{-1}\sum_{k=1}^{N} c_{1(k)}c_{2(k)}^* = N^{-1}\sum_{k=1}^{N} C_{1(k)}(t)C_{2(k)}^*(t)\exp\left(\frac{-iE_{12(k)}t}{\hbar}\right)$$

$$E_{12(k)} \equiv E_{1(k)} - E_{2(k)}$$

If E_{12} are different in the ensemble, the off-diagonal element averages to zero after some time as oscillations run out of phase

Note: if your measurable quantity off-diagonal elements are not zero, the measurement may also oscillate for some time!

10

Von Neumann equation

$$\boxed{\rho_{nm}(t) = c_n(t)c_m^*(t)}$$

Time-dependence of ρ

Let the wave function be: $\Psi(t) = \sum_n c_n(t)\psi_n(\mathbf{r})$

$$c_n(t) = C_n(t)e^{-iH_{nn}t/\hbar}$$

Schrödinger: $i\hbar\sum_n \frac{\partial c_n}{\partial t}\psi_n = \hat{H}\sum_n c_n\psi_n$

$$i\hbar\frac{\partial \Psi}{\partial t} = \hat{H}\Psi$$

Use $\langle\psi_m|\cdots\rangle$

$$\rho_{mn} = \rho_{nm}^*$$

$$\sum_n \langle\psi_m|\psi_n\rangle\frac{\partial c_n}{\partial t} = -\frac{i}{\hbar}\sum_n c_n\langle\psi_m|\hat{H}|\psi_n\rangle$$

Time-dependence: $\frac{\partial c_m}{\partial t} = -\frac{i}{\hbar}\sum_k H_{mk}c_k$ Note that *k* runs over all including *m*

On the other hand:

$$\frac{\partial\rho_{nm}}{\partial t} = \frac{\partial(c_n c_m^*)}{\partial t} = c_n\frac{\partial c_m^*}{\partial t} + c_m^*\frac{\partial c_n}{\partial t} = (i/\hbar)\left(c_n\sum_k H_{mk}^* c_k^* - c_m^*\sum_k H_{nk}c_k\right)$$

$$= (i/\hbar)\sum_k (H_{km}\rho_{nk} - H_{nk}\rho_{km})$$

Used: $H_{mk}^* = H_{km}$

$$= (i/\hbar)\sum_k \{\rho_{nk}H_{km} - H_{nk}\rho_{km}\}$$

(H is Hermitian operator)

Von Neumann equation

$$\boxed{\rho_{nm}(t) = c_n(t)c_m^*(t)}$$

Time-dependence of ρ:

Have: $\frac{\partial\rho_{nm}}{\partial t} = (i/\hbar)\sum_k \{\rho_{nk}H_{km} - H_{nk}\rho_{km}\}$

Can rewrite in terms of matrix products:

$$\frac{\partial\rho_{nm}}{\partial t} = (i/\hbar)\{[\boldsymbol{\rho}\mathbf{H}]_{nm} - [\mathbf{H}\boldsymbol{\rho}]_{nm}\} \equiv (i/\hbar)[\boldsymbol{\rho},\mathbf{H}]_{nm}$$

Commutator:
[A,B] ≡ AB - BA

Or for all elements as a single matrix expression:

$$\frac{\partial\boldsymbol{\rho}}{\partial t} = (i/\hbar)[\boldsymbol{\rho},\mathbf{H}]$$

For ensemble average of N systems

$$\boxed{\frac{\partial\overline{\boldsymbol{\rho}}}{\partial t} = (i/\hbar)[\overline{\boldsymbol{\rho}},\mathbf{H}]}$$

von Neumann equation (1927)

John von Neumann

1903-1957

(It is also known as the Liouville equation because of its parallel to Liouville's classical statistical mechanical theorem on the density of dynamic variables in phase space)

Von Neumann equation

Example:
Three-state system:

$$\Psi(t) = \sum_n c_n(t)\psi_n(\mathbf{r})$$

$$c_n(t) = C_n(t)e^{-iH_{nn}t/\hbar}$$

$$\frac{\partial \boldsymbol{\rho}}{\partial t} = (i/\hbar)[\boldsymbol{\rho}, \mathbf{H}] \qquad \boxed{\rho_{nm}(t) = c_n(t)c_m^*(t)}$$

$$\frac{\partial}{\partial t}\begin{pmatrix}\rho_{11} & \rho_{12} & \rho_{13}\\ \rho_{21} & \rho_{22} & \rho_{23}\\ \rho_{31} & \rho_{32} & \rho_{33}\end{pmatrix} = \frac{i}{\hbar}\left[\begin{pmatrix}\rho_{11} & \rho_{12} & \rho_{13}\\ \rho_{21} & \rho_{22} & \rho_{23}\\ \rho_{31} & \rho_{32} & \rho_{33}\end{pmatrix}\begin{pmatrix}H_{11} & H_{12} & H_{13}\\ H_{21} & H_{22} & H_{23}\\ H_{31} & H_{32} & H_{33}\end{pmatrix} - \begin{pmatrix}H_{11} & H_{12} & H_{13}\\ H_{21} & H_{22} & H_{23}\\ H_{31} & H_{32} & H_{33}\end{pmatrix}\begin{pmatrix}\rho_{11} & \rho_{12} & \rho_{13}\\ \rho_{21} & \rho_{22} & \rho_{23}\\ \rho_{31} & \rho_{32} & \rho_{33}\end{pmatrix}\right]$$

$$\partial\rho_{11}/\partial t = (i/\hbar)(\underline{\rho_{11}H_{11}} + \rho_{12}H_{21} + \rho_{13}H_{31} - \underline{\rho_{11}H_{11}} - \rho_{21}H_{12} - \rho_{31}H_{13})$$

$$= (i/\hbar)(\rho_{12}H_{21} - \rho_{21}H_{12} + \rho_{13}H_{31} - \rho_{31}H_{13}).$$

What does $\rho_{11}(t)$ represent?

13

Von Neumann equation

Example: Three-state system: $\boxed{\rho_{nm}(t) = c_n(t)c_m^*(t)}$

populations

$$\partial\rho_{11}/\partial t = (i/\hbar)(\boxed{\rho_{12}H_{21} - \rho_{21}H_{12}} + \rho_{13}H_{31} - \rho_{31}H_{13})$$
$$\partial\rho_{22}/\partial t = (i/\hbar)(\boxed{\rho_{21}H_{12} - \rho_{12}H_{21}} + \rho_{23}H_{32} - \rho_{32}H_{23})$$
$$\partial\rho_{33}/\partial t = (i/\hbar)(\rho_{31}H_{13} - \rho_{13}H_{31} + \rho_{32}H_{23} - \rho_{23}H_{32})$$

Off-diagonal

$$\partial\rho_{12}/\partial t = (i/\hbar)[\boxed{(\rho_{22} - \rho_{11})H_{21} + \rho_{21}(H_{11} - H_{22})} + \rho_{23}H_{31} - \rho_{31}H_{23}]$$
$$\partial\rho_{21}/\partial t = (i/\hbar)[\boxed{(\rho_{22} - \rho_{11})H_{21} + \rho_{21}(H_{11} - H_{22})} + \rho_{23}H_{31} - \rho_{31}H_{23}]$$
$$\partial\rho_{13}/\partial t = (i/\hbar)[(\rho_{11} - \rho_{33})H_{13} + \rho_{13}(H_{33} - H_{11}) + \rho_{12}H_{23} - \rho_{23}H_{12}]$$
$$\partial\rho_{31}/\partial t = (i/\hbar)[(\rho_{33} - \rho_{11})H_{31} + \rho_{31}(H_{11} - H_{33}) + \rho_{32}H_{21} - \rho_{21}H_{32}]$$
$$\partial\rho_{23}/\partial t = (i/\hbar)[(\rho_{22} - \rho_{33})H_{23} + \rho_{23}(H_{33} - H_{22}) + \rho_{21}H_{13} - \rho_{13}H_{21}]$$
$$\partial\rho_{32}/\partial t = (i/\hbar)[(\rho_{33} - \rho_{22})H_{32} + \rho_{32}(H_{22} - H_{33}) + \rho_{31}H_{12} - \rho_{12}H_{31}]$$

How do populations evolve?

- Need off-diagonal elements to build-up populations! is not instantaneous
- Consistent with what we had for 2-state system (red boxes above):

$$\partial C_1/\partial t = -(i/\hbar)H_{12}\exp(iE_{12}t/\hbar)C_2$$
$$\partial C_2/\partial t = -(i/\hbar)H_{21}\exp(iE_{21}t/\hbar)C_1$$

but now it depends on off-diagonal elements, not on populations directly!

Note: populations are balanced: $\partial\overline{\rho_{mm}}/\partial t = -\sum_{n\neq m}\partial\overline{\rho_{nn}}/\partial t$

14

Von Neumann equation

Example: Three-state system:

$\boxed{\rho_{nm}(t) = c_n(t)c_m^*(t)}$

populations

$$\partial\rho_{11}/\partial t = (i/\hbar)(\rho_{12}H_{21} - \rho_{21}H_{12} + \rho_{13}H_{31} - \rho_{31}H_{13})$$
$$\partial\rho_{22}/\partial t = (i/\hbar)(\rho_{21}H_{12} - \rho_{12}H_{21} + \rho_{23}H_{32} - \rho_{32}H_{23})$$
$$\partial\rho_{33}/\partial t = (i/\hbar)(\rho_{31}H_{13} - \rho_{13}H_{31} + \rho_{32}H_{23} - \rho_{23}H_{32})$$

Off-diagonal

$$\partial\rho_{12}/\partial t = (i/\hbar)[(\rho_{22} - \rho_{11})H_{21} + \rho_{21}(H_{11} - H_{22}) + \rho_{23}H_{31} - \rho_{31}H_{23}]$$
$$\partial\rho_{21}/\partial t = (i/\hbar)[(\rho_{22} - \rho_{11})H_{21} + \rho_{21}(H_{11} - H_{22}) + \rho_{23}H_{31} - \rho_{31}H_{23}]$$
$$\partial\rho_{13}/\partial t = (i/\hbar)[(\rho_{11} - \rho_{33})H_{13} + \rho_{13}(H_{33} - H_{11}) + \rho_{12}H_{23} - \rho_{23}H_{12}]$$
$$\partial\rho_{31}/\partial t = (i/\hbar)[(\rho_{33} - \rho_{11})H_{31} + \rho_{31}(H_{11} - H_{33}) + \rho_{32}H_{21} - \rho_{21}H_{32}]$$
$$\partial\rho_{23}/\partial t = (i/\hbar)[(\rho_{22} - \rho_{33})H_{23} + \rho_{23}(H_{33} - H_{22}) + \rho_{21}H_{13} - \rho_{13}H_{21}]$$
$$\partial\rho_{32}/\partial t = (i/\hbar)[(\rho_{33} - \rho_{22})H_{32} + \rho_{32}(H_{22} - H_{33}) + \rho_{31}H_{12} - \rho_{12}H_{31}]$$

What if $H_{12} = H_{13} = H_{23} = 0$? $\quad H_{mn} \equiv \langle \psi_m | \hat{H} | \psi_n \rangle$

- populations cannot change
- off-diagonals depend only on energy differences between states and itself

$$\partial\rho_{32}/\partial t = (i/\hbar)(H_{22} - H_{33})\rho_{32} \Rightarrow \rho_{32} = \rho_{32}^{(0)} \exp\left[i(H_{22} - H_{33})t/\hbar\right]$$

beats with $(H_{22}-H_{33})/h$ frequency

Once coherence is lost, off-diagonal elements should go to zero for the ensemble!
Need external perturbation to initiate coherence!

15

EXTRA: Von Neumann equation

Example: Two-state system:

$\boxed{\rho_{nm}(t) = c_n(t)c_m^*(t)}$

$$c_1 = C_1 e^{-iH_{11}t/\hbar}$$
$$c_2 = C_2 e^{-iH_{22}t/\hbar}$$

populations

$$\partial\rho_{11}/\partial t = (i/\hbar)(\boxed{\rho_{12}H_{21} - \rho_{21}H_{12}})$$
$$\partial\rho_{22}/\partial t = (i/\hbar)(\boxed{\rho_{21}H_{12} - \rho_{12}H_{21}})$$

Off-diagonal

$$\partial\rho_{12}/\partial t = (i/\hbar)[\boxed{(\rho_{22} - \rho_{11})H_{21} + \rho_{21}(H_{11} - H_{22})}$$
$$\partial\rho_{21}/\partial t = (i/\hbar)[\boxed{(\rho_{22} - \rho_{11})H_{21} + \rho_{21}(H_{11} - H_{22})}$$

$$\rho_{nm} = c_n c_m^* = \begin{pmatrix} C_1 C_1^* & C_1 C_2^* e^{-i(H_{11}-H_{12})t/\hbar} \\ C_2 C_1^* e^{i(H_{11}-H_{12})t/\hbar} & C_2 C_2^* \end{pmatrix}$$

Initial conditions $C_1 = 1$, $C_2 = 0$, find C_2

Can derive:

$$\partial C_1/\partial t = -(i/\hbar)H_{12}\exp(iE_{12}t/\hbar)C_2$$
$$\partial C_2/\partial t = -(i/\hbar)H_{21}\exp(iE_{21}t/\hbar)C_1$$

16

The Stochastic Liouville Equation

Assume interaction with surroundings:
→ Energy transfer to surrounding should bring system into Boltzmann equilibrium.

In principle, can include surrounding into system and solve: $\frac{\partial \boldsymbol{\rho}}{\partial t} = (i/h)[\boldsymbol{\rho}, \mathbf{H}]$
- But that will produce enormous matrices, virtually impossible

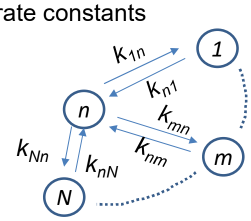
Instead: consider ensemble of individual systems, account for surrounding in a statistical way – respective matrix is called the ***reduced*** density matrix.

Diagonal elements of the reduced density matrix correspond to population that should evolve toward equilibrium values satisfying Boltzmann distribution:

$$\overline{\rho_{nn}^{o}} = Z^{-1}\exp(-E_n/k_B T)$$

Classically evolution kinetics will be described by a set of rate constants for interconversion between state *n* and all other states.
Assuming first-order kinetics: k_{nm} → from *m* to *n*

$$\{\partial\overline{\rho_{nn}}/\partial t\}_{stochastic} = \sum_{m \neq n}(k_{nm}\overline{\rho_{mm}} - k_{mn}\overline{\rho_{nn}})$$



k_{ij} – microscopical classical rate constants *j*→*i*

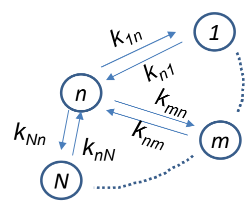
$_{stochastic}$ – *relaxations depend on random fluctuations of environment, no quantum oscillations*

17

The Stochastic Liouville Equation

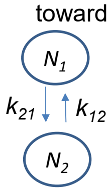
Assume interaction with surrounding:

$$\{\partial\overline{\rho_{nn}}/\partial t\}_{stochastic} = \sum_{m \neq n}(k_{nm}\overline{\rho_{mm}} - k_{mn}\overline{\rho_{nn}})$$



Notes:
- evolution above does not depend on the off-diagonal elements of ρ
- due to Boltzmann equilibrium → $\frac{k_{mn}}{k_{nm}} = \exp\left(-\frac{E_{nm}}{kT}\right)$

Consider: two-level system → populations must evolve with the same rate $1/T_1$ toward Boltzmann equilibrium



$$\begin{cases} \frac{dN_1}{dt} = -k_{21}N_1 + k_{12}N_2 \\ \frac{dN_2}{dt} = +k_{21}N_1 - k_{12}N_2 \end{cases} \Rightarrow \begin{cases} \frac{d\left(N_1 - N_1^0\right)}{dt} = -\frac{1}{T_1}\left(N_1 - N_1^0\right) \\ \frac{d\left(N_2 - N_2^0\right)}{dt} = -\frac{1}{T_1}\left(N_2 - N_2^0\right) \end{cases} \qquad \frac{N_1^0}{N_2^0} = \frac{k_{21}}{k_{12}} \qquad \frac{1}{T_1} = k_{12} + k_{21}$$

Or in terms of density matrix:

Boltzmann-equilibrated population

$$\left[\partial\left(\bar{\rho}_{nn} - \bar{\rho}_{nn}^0\right)/\partial t\right]_{stochastic} = -\frac{1}{T_1}\left(\bar{\rho}_{nn} - \bar{\rho}_{nn}^0\right) \qquad \frac{1}{T_1} = k_{12} + k_{21}$$

18

The Stochastic Liouville Equation

Two-level system → populations must evolve with the same rate $1/T_1$ toward Boltzmann equilibrium

$$\boxed{\left[\partial\left(\overline{\rho}_{nn}-\overline{\rho}^{0}_{nn}\right)/\partial t\right]_{stochastic} = -\frac{1}{T_1}\left(\overline{\rho}_{nn}-\overline{\rho}^{0}_{nn}\right)} \qquad \boxed{\frac{1}{T_1} = k_{12}+k_{21}}$$

$$\overline{\rho}_{11} \underset{k_{12}}{\overset{k_{21}}{\rightleftarrows}} \overline{\rho}_{22}$$

T_1 is called *longitudinal relaxation time* or *spin-lattice relaxation time* in NMR and EPR spectroscopies and relates to nuclear relaxation of the z-component of spin)
In optical spectroscopy, T_1 is called "population" decay time, or just T1-time.

Off-diagonal terms should go to 0 at equilibrium if ρ is defined in *stationary* basis:

- Stochastic fluctuations of diagonal elements will lead to loss of coherence and ensemble-averaged values of off-diagonal elements go to 0: c_k will acquire random phase ζ. Can show that T_1 decay of diagonal elements leads to decay of off-diagonal elements that is twice faster ($T_1/2$).
- Inhomogeneity in the energies of individual systems → oscillations get out of phase. This is called <u>*pure dephasing*</u> and lifetime is denoted as T_2^* *(T2*-time).*

$$\boxed{\{\partial\overline{\rho_{nm}}/\partial t\}_{stochastic} = -\frac{1}{T_2}\overline{\rho_{nm}}} \quad \text{for } m\neq n\text{, and} \quad \boxed{1/T_2 \approx 1/T_2^* + 1/2T_1}$$

T_2 – the *tranansverse relaxation time,* or just T2-time

19

The Stochastic Liouville Equation

Dynamics of diagonal and off-diagonal elements:

$$\partial\overline{\rho_{nm}}/\partial t = (i/\hbar)[\,\overline{\boldsymbol{\rho}},\mathbf{H}]_{nm} + \sum_{j,k} R_{nm.jk}\overline{\rho_{jk}}$$

$$\{\partial\overline{\rho_{nm}}/\partial t\}_{stochastic} = -\frac{1}{T_2}\overline{\rho_{nm}}$$

Stochastic Liouville Equation:

$$\boxed{\partial\overline{\boldsymbol{\rho}}/\partial t = (i/\hbar)[\overline{\boldsymbol{\rho}},\mathbf{H}] + \mathbf{R}\overline{\boldsymbol{\rho}}}$$

von Neumann for system
+ Rρ for bath

R – <u>*relaxation matrix.*</u> = set of rate constants for stochastic processes.

R is in fact a fourth-order tensor, while ρ is second order tensor (or matrix).

$$\left(\mathbf{R}\overline{\boldsymbol{\rho}}\right)_{nm} = \sum_{i,k} R_{nmjk}\overline{\boldsymbol{\rho}}_{jk}$$

20

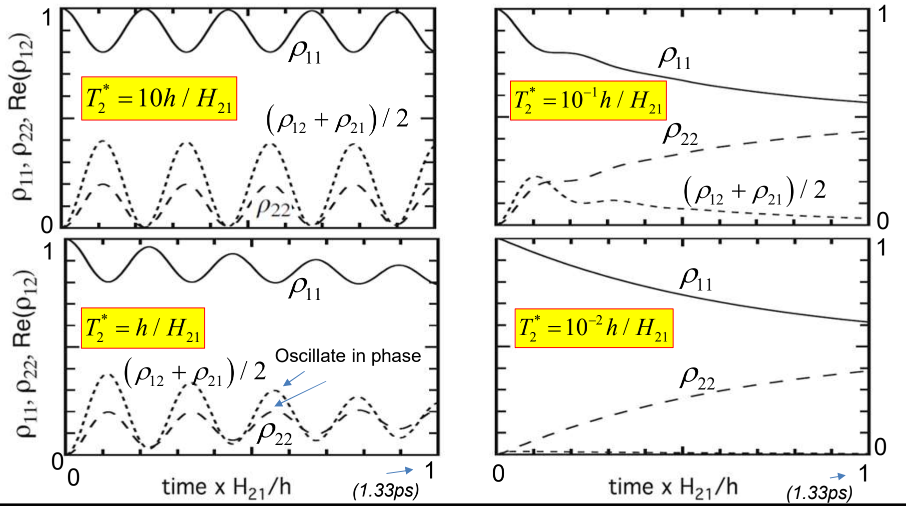
The Stochastic Effects

$\partial\overline{\boldsymbol{\rho}}/\partial t = (i/h)[\overline{\boldsymbol{\rho}}, \mathbf{H}] + \mathbf{R}\overline{\boldsymbol{\rho}}$

Example: 2 states.

$H_{21} = E_{12}/4$; $T_1 = 10^4 h/H_{21}$; E_{12}=100cm^{-1} (H_{11}-H_{22}); H_{21}=25cm^{-1}; period = $h/2\Omega$ = 0.3ps

Long T_1: Negligible transition between states, coherence is lost by pure dephasing. thermal equilibrium not achieved within this simulation time, dephasing traps in both states with equal probability



Note: T_1 is very long, no Boiltzmann eq. reached…

21

The Stochastic Effects

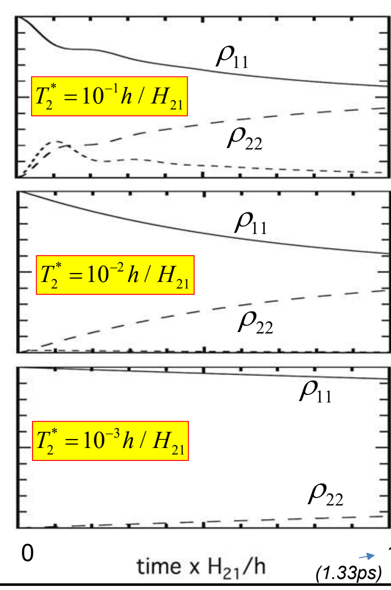
$\partial\overline{\boldsymbol{\rho}}/\partial t = (i/h)[\overline{\boldsymbol{\rho}}, \mathbf{H}] + \mathbf{R}\overline{\boldsymbol{\rho}}$

Example: 2 states, continued

When T_2^* is very short (<<|h/H_{12}|) off-diagonals are quenched very rapidly and interconversion of states 1 and 2 are suppressed.

Populations evolve toward equal values: the stochastic decay of ρ_{12} and ρ_{21} caused by pure dephasing is equally likely to trap the quantum system in either state. Thermal (Boltzmann) equilibrium would be attained only at times exceeding T1 ~ 10 ns

Note: as dephasing becomes faster the energy transfer becomes slower!



22

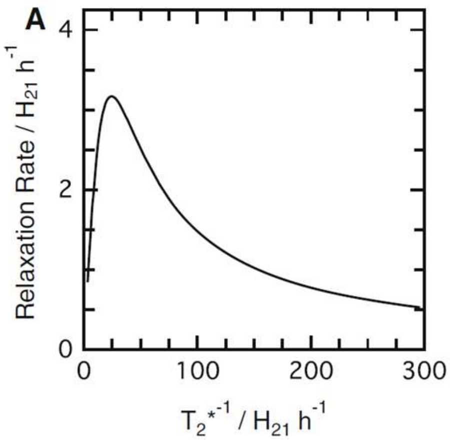
The Stochastic Effects

Example: 2 state. Continued…

Relaxation rate of ensemble as a function of $1/T_2^*$

Relaxation rate is 1/T, where T is time at which population of state 1 falls from 1 to 0.816=(1-0.5/e)

Faster transfer (relaxation) at optimal dephasing!



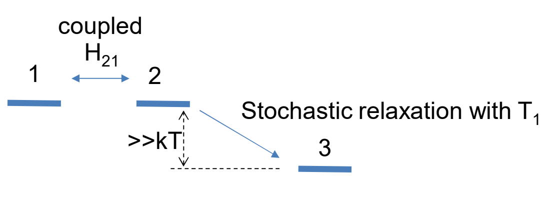
23

The Stochastic Effects

$\boxed{\partial\overline{\boldsymbol{\rho}}/\partial t = (i/h)[\overline{\boldsymbol{\rho}}, \mathbf{H}] + \mathbf{R}\overline{\boldsymbol{\rho}}}$

Example: 3 states:
- States 1 & 2 are at the same energy (E_{21}=0) and interact ($H_{21}\neq0$), interconversion is driven by QM coupling only
- State 2 can stochastically relax to state 3 at lower energy – resonance energy transfer
- Assume pure dephasing T_2^* = 1 ns → is very long

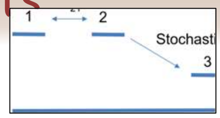
i.e. $R_{11}=R_{22}=0$; $H_{13}=0$



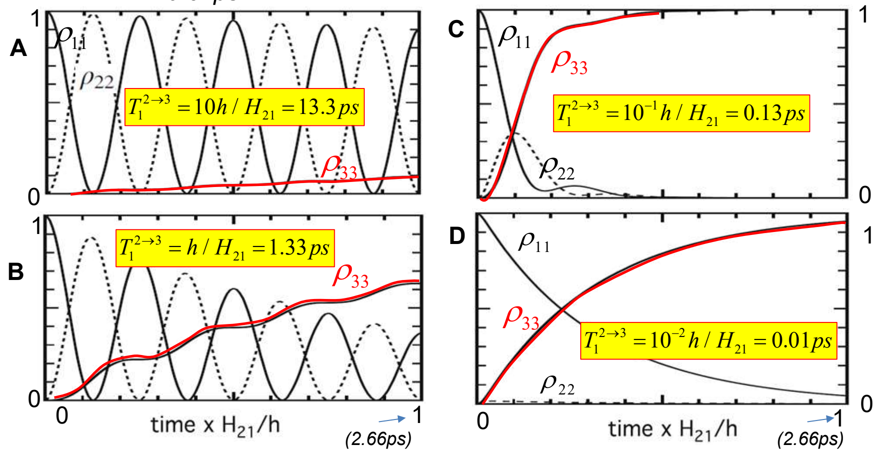
Ensemble starts in state 1
How rapidly will it appear in state 3?

24

The Stochastic Effects



Example: 3 states: $E_{21}=0$, $E_{32}=-40|H_{21}| >> kT$
T_2^* =750h/H21=10^3ps; E_{23}=1000cm^{-1}; H_{21}=25cm^{-1};
period = *h/2Ω = 0.67ps*



Notice: as T_1 becomes smaller the energy transfers to 3 faster (A-B-C).
BUT! For even faster T_1 the formation of state 3 slows down!!!
This is contrary to expectations from FRET theory!

25

The Stochastic Effects

Example: 3 state. Contined…
T_2^* =750h/H21=10^3ps
Oscillation frequency = 2($H_{21}h^{-1}$)

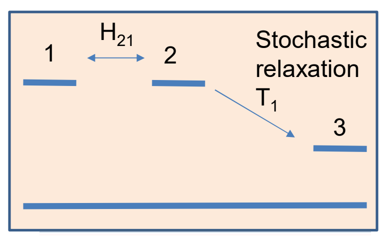
Relaxation rate of ensemble as a function of $1/T_1$

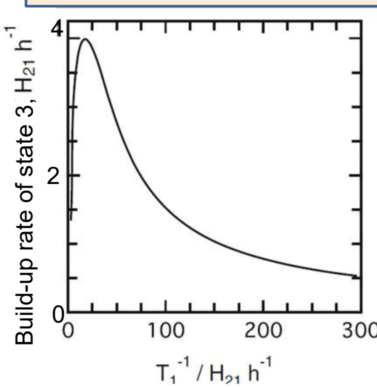
1/T, where T is time at which population of state 3 rises from 0 to 0.634=(1-e).

When T_1 is very short, i.e. energy transfer from 2→3, is very fast, the energy transfer slows down!!??

…This is the consequence of stochastic Liouville equation….

$$\partial\overline{\boldsymbol{\rho}}/\partial t = (i/h)[\overline{\boldsymbol{\rho}}, \mathbf{H}] + \mathbf{R}\overline{\boldsymbol{\rho}}$$

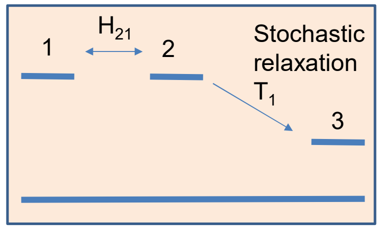




26

The Stochastic Effects

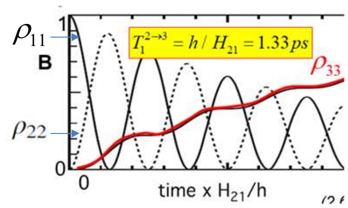
Example: 3 state. Contined…
When energy transfer from 2→3, is very fast, the energy transfer 1→3 slows down!!??



1. Rate of transfer to (3) depends on:
   - Transfer rate (2)→(3), i.e. $1/T_1$
   - population of (2):

$$C_2(t) = -i(H_{21}/\Omega)\sin(\Omega t/h)\exp(-iE_{21}t/2h)$$

$$|C_2(t)|^2 = \left|\frac{H_{21}}{\Omega}\right|^2 \sin^2\left(\frac{\Omega t}{\hbar}\right)$$



2. Transfer (2)→(3) occurs at time ~T_1, when population is:

$$|C_2(t)|^2 \propto \sin^2\left(\frac{\Omega}{\hbar}T_1\right) \approx T_1^2 \qquad \text{(for small } T_1\text{)}$$

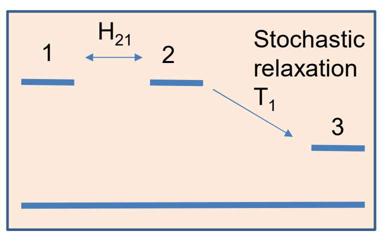
3. Rate of transfer to (3) is thus $Rate(3) = Populatin(2) \times 1/T_1 \propto T_1^2 \times T_1^{-1} = T_1$

I.e. rate of population (3) formation becomes *smaller* (slower!) when time T_1 Becomes *smaller* (i.e. faster!)

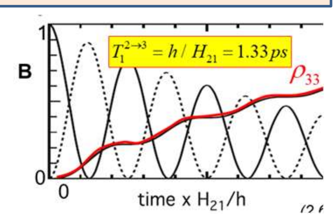
27

The Stochastic Effects

Example: 3 state. Contined…
When energy transfer from 2→3, is very fast, the energy transfer 1→3 slows down!!??



In other words:
- Stochastic process 2→3 constantly *tests* the state of quantum system 1+2 with time interval ~T_1.
- Each "test" brings the system to a *definite* state (1), (2) or (3).
- At short times probability to find system in state (2) is small (~T_1^2) and system is thus collapsed into definite state (1) and we are at "zero" time on the graph again!



Quantum Zeno effect: particle's time evolution is arrested by measuring it frequently enough with respect to some chosen measurement setting

This is similar to the "quantum arrow" question – will it hit you faster when you look at it, or when you don't?

28